Sinyin Chang

sinyinchang@gmail.com | linkedin.com/in/sinyinchang | sinyinchang.github.io

EDUCATION

Johns Hopkins University (JHU), Baltimore, MD — Expected May 2027
Master of Science in Engineering (MSE), Computer Science

National Taiwan University (NTU), Taipei, Taiwan — Jun 2025
Bachelor of Science in Computer Science and Information Engineering

EXPERIENCE

Miraivant Advisory, Rockville, MD — Oct 2026 – Present
AI Research Intern

- Conducting technical research on healthcare referral automation platforms to inform the design and development of PatientFlow AI, an AI-powered platform for streamlining patient referral workflows and decision support.

Washington University in St. Louis, St. Louis, MO — Jun 2024 – Aug 2024
Research Intern

- Led a first-author project developing an adaptive input sanitization framework for legacy medical systems using extended Berkeley Packet Filter (eBPF), enabling runtime deployment of kernel-level defenses without service downtime while achieving <19 ms patch loading time.
- Developed an automated lighting-control system for the Open Platform for Cyber-Physical Systems (OPCPS) by implementing a custom network-based interface connecting software with physical lighting hardware.

National Taiwan University, Taipei, Taiwan — Jan 2024 – Jun 2024
Research Intern

- Collaborated with PhD students on gait-analysis research, evaluating monocular depth-estimation models and integrating open-source computer-vision pipelines to explore 3D depth recovery from 2D video.

PROJECTS

Modeling Social Pressure in Multi-Agent LLM — May 2026

- Built a Python-based multi-agent LLM simulation and evaluation framework using model APIs and prompt engineering to analyze cooperative behavior across 5 LLM families and 3 experimental conditions.

CompareAI: User-Centered AI Tool Comparison Platform Environments — May 2026

- Designed an AI tool comparison platform in Figma through contextual inquiry, participatory design, and usability testing; usability tasks, including tool comparison, achieved 75–100% completion in ~20 seconds.

I-CALM: Incentivizing Confidence-Aware Abstention for LLM Hallucination Mitigation — Mar 2026

- Engineered a confidence-aware LLM framework using model APIs and prompt engineering; evaluated 5 LLM families across 3 QA datasets, reducing false-answer rate from 52.3% to 34.2% on GPT-5 mini/PopQA.

Full-Stack Meeting Coordination Platform — Jun 2024

- Implemented frontend and backend components for a meeting coordination platform using Python, JavaScript, HTML, CSS, and MongoDB, contributing to an integrated system that streamlined meeting coordination.

PUBLICATIONS

1. Haotian Zong, Binze Li, Yufei Long, Sinyin Chang, Jialong Wu, Gillian K Hadfield — Apr 2026
   I-CALM: Incentivizing Confidence-Aware Abstention for LLM Hallucination Mitigation
   Epistemic Intelligence in Machine Learning Workshop, co-located with ICML 2026

2. Sinyin Chang, Ao Li, Evin Jaff, Yuanhaur Chang, Jinwen Wang, Ning Zhang, Hsu-Chun Hsiao — Dec 2024
   AdapSan: Adaptive Input Sanitization in Medical Systems with eBPF
   ACM Workshop on Adaptive and Autonomous Cyber Defense, 2024

3. Ao Li, Sinyin Chang, Guorui Li, Yuanhaur Chang, Nathan Fisher, Thidapat Chantem — Nov 2023
   Software and Behavior Diversification for Swarm Robotics Systems
   ACM Workshop on Moving Target Defense, 2023

TECHNICAL SKILLS

Programming & Tools: Python, C/C++, JavaScript, SQL, HTML, CSS, React, Node.js, MongoDB, Git, Docker
AI/ML: LLMs, Prompt Engineering, Multi-Agent Systems, LLM APIs, Agent/Model Evaluation